Can app metadata identify highly rated apps?

ENDTERM / MACHINE LEARNING ALGORITHMS

Seven inputs. Eight models.
One frozen test set in the new experiment.

SVM selected by training CV. Test F1 0.887; class-0 recall 8.0%.

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Continuity is the unresolved requirement

October Midterm

2,029 apps
Original private snapshot and split were lost.
Public aggregates survive.

November Endterm

2,179 real API apps
1,743 train / 436 test
Raw data and split now retained privately.

New snapshot: no claim of improvement on the original holdout

Every choice is made inside training data

Inputs	Price, size, languages, release year, update recency, genre, age category
Outer CV	Five shared stratified folds compare all eight methods
Inner CV	Three folds tune forest and boosting inside each outer training fold
Final test	Evaluate after CV selection; keep default decision thresholds

One comparison includes every baseline

Model	CV F1 +/- SD	Test F1	AUC	Bal. acc.
Dummy	0.888 +/- 0.001	0.888	0.500	0.500
Decision Tree	0.864 +/- 0.017	0.875	0.577	0.569
KNN	0.889 +/- 0.004	0.868	0.651	0.535
SVM *	**0.893 +/- 0.003**	**0.887**	**0.643**	**0.531**
Random Forest	0.888 +/- 0.002	0.887	0.698	0.572
Gradient Boosting	0.890 +/- 0.002	0.891	0.644	0.521
SVM + PCA	0.892 +/- 0.002	0.889	0.619	0.528
MLP	0.875 +/- 0.007	0.864	0.645	0.600

* Selected by CV. Spread is sample SD. All test rows are identical.

Small, explicit searches control complexity

Random Forest

max_depth: None
max_features: sqrt
min_samples_leaf: 2

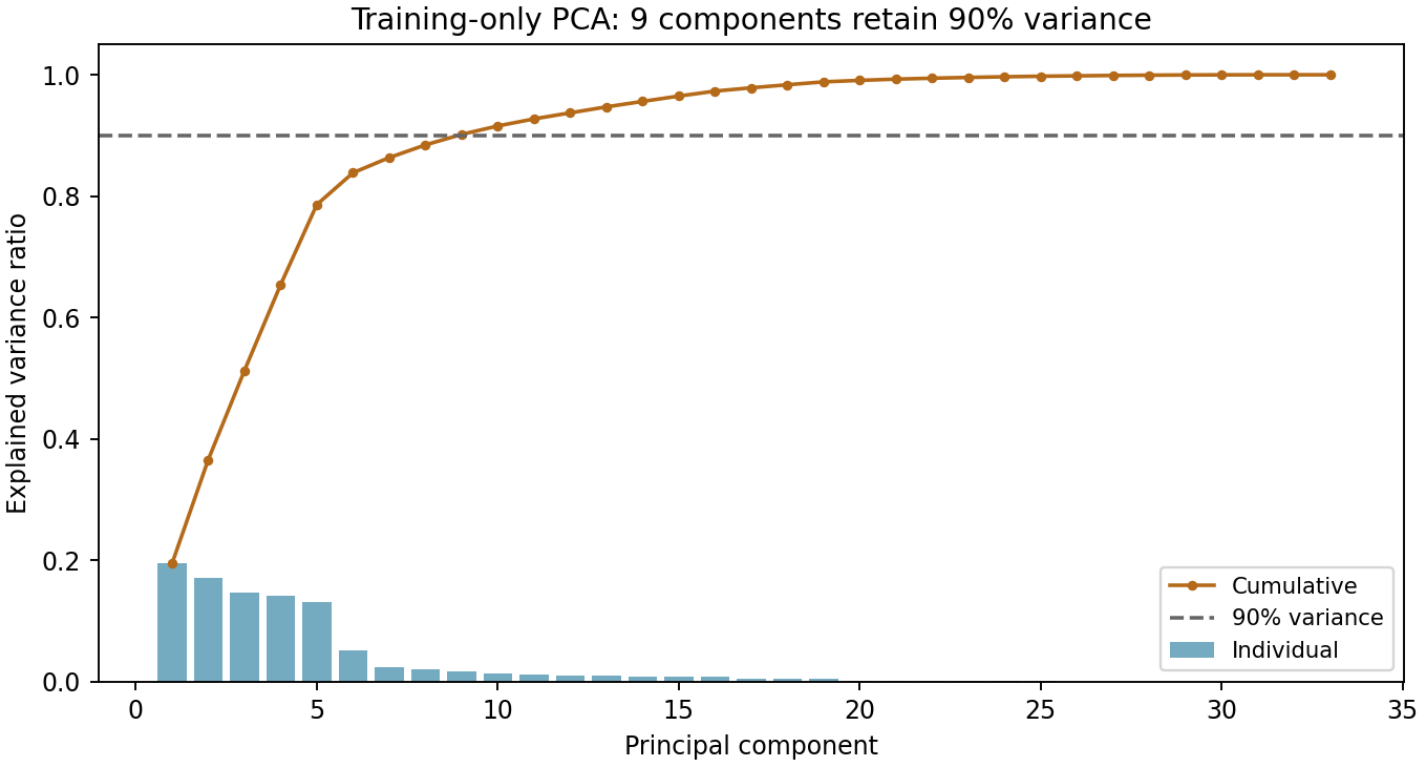
CV F1 0.888
Test F1 0.887

Gradient Boosting

learning_rate: 0.03
max_depth: 2
n_estimators: 75

CV F1 0.890
Test F1 0.891

PCA describes variation, then tests its value



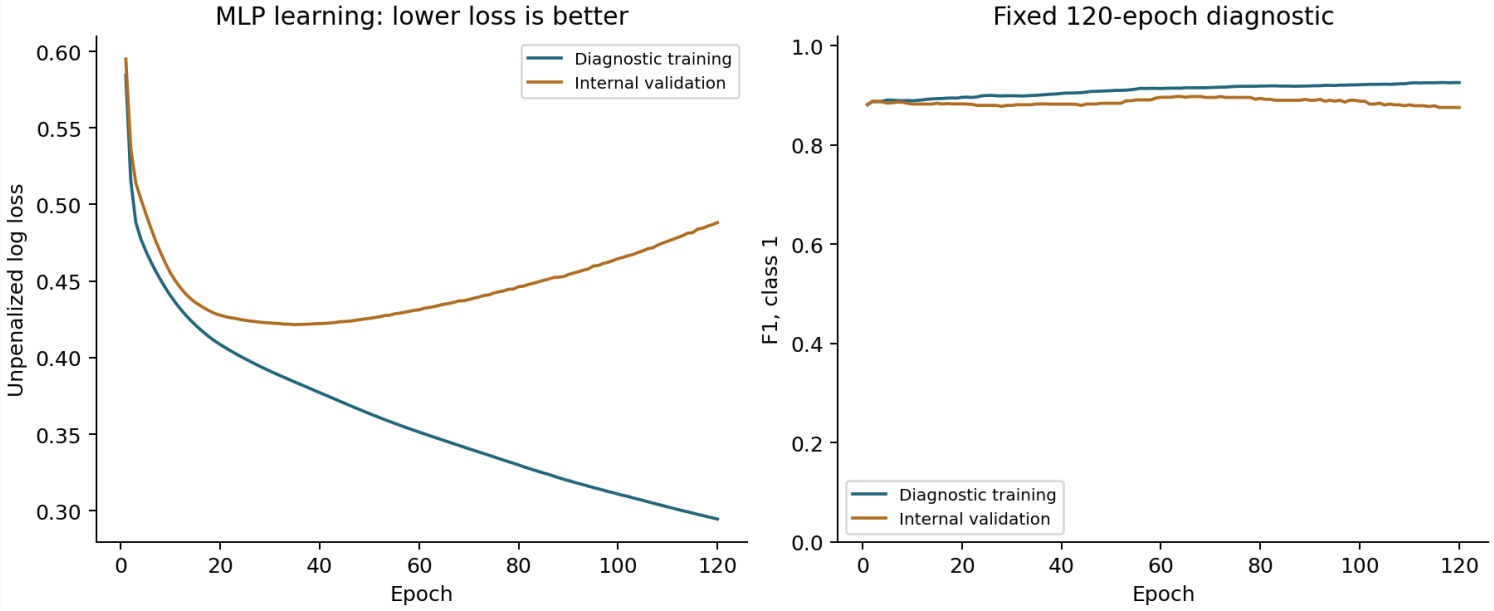
9 components retain 90% variance.

First two: 36.6%.

PCA - SVM CV F1: -0.0007

Numeric inputs standardized; one-hot categories stay 0/1. No claim of natural clusters.

The MLP learning check reveals overfitting



32 + 16 hidden units
ReLU / Adam / 120 epochs

Validation-loss minimum:
epoch 35

MLP CV F1: 0.875
Earlier best: SVM
0.893

Error patterns matter more than a high F1

SVM on the holdout

Lower-rated detected	**7 / 88**
False positives	**81**
False negatives	**6**

All models: matrices + actual errors

Boundary cases and coarse metadata describe associations.

They do not explain why users assigned ratings.

Conclusions and a testable Final plan

Metadata offers limited discrimination

Query sampling, imbalance, developer overlap and snapshot drift limit generalization.

Final: permission-checked descriptions linked to the same apps. Compare text-only, metadata-only and combined models using training-fold TF-IDF.

Codex assistance disclosed. Real contributions and individual explanations belong to the students.